

نموذج إجابة

الامتحان دور ثاني 2024

Group 1

false	2	Tesla	1
true	4	true	3
Fleming left hand rule	6	false	5
B) $\frac{1}{2} B$	8	$F=0.1 \times 20 \times 1 = 2N$	7
$B = \frac{\mu NI}{2r} = \frac{4\pi \times 10^{-7} \times 1 \times 2}{2 \times 4 \times 10^{-2}} = \pi \times 10^{-5}$	10	B) 2B	9

Group 2

C) south – North	12	a) Increases to double	11
$emf_2 = \frac{-N \Delta \Phi_m}{\Delta t}$	14	$M = \frac{emf \Delta t}{\Delta I}$	13
Lenz rule “determine the direction of induced current in a coil”	16	Fleming right hand rule “determine the direction of induced current in the wire”	15
Self-induction	18	Eddy currents	17
false	20	true	19

Group 3

X-Rays	22	Inversely	21
The name: Coolidge tube	24	10^{-3}	23
b) $\frac{1}{2} m v^2$	26	Use: Production of X-rays	25
c.)	28	c) 1:2	27
true	30	False	29

Group4

Analog electronics	32	Greater than	31
Equal	34	Greater than	33
OR	36	Not	35
0	38	0	37
reverse high resistance	40	Forward small resistance	39

Group5

Mention two properties of LASER	41
Monochromaticity - collimation - coherence	42
ملغي	44
ملغي	43
$\tau_{max} = BIAN = 10 \times 1 \times (0.08 \times 0.05) \times 100 = 4 \text{N.m}$	45
	46
$I_E = I_B + I_C$	47
$I_C = I_E + I_B = 100 - 2 = 98 \text{mA} = 98 \times 10^{-3} \text{A}$	48
Step-up	49
Because $V_s > V_p$	59